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# DRAPE: Doppler-Robust Acoustic Pose Estimation of Rigid Bodies with Three Degrees of Freedom Using FMCW-TDoA

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***Abstract*— This paper presents a high-precision acoustic positioning system with three degrees of freedom (3-DoF) for indoor mobile robots, based on a microphone array mounted on the robot and multiple acoustic anchors deployed in the environment. The system first estimates time-difference-of-arrival (TDoA) measurements using an improved FMCW-based ranging scheme that is specifically designed for indoor acoustic propagation conditions. To address the adverse effects introduced by robot motion, a Doppler-robust signal processing strategy is incorporated to effectively eliminate motion-induced Doppler bias without increasing signal bandwidth. Building upon the obtained TDoA measurements, a rigid-body-constrained pose estimation framework is then employed to jointly process multi-channel microphone observations, enabling robust and consistent estimation of both position and yaw angle. Comprehensive experiments conducted in a real indoor environment demonstrate that the proposed method achieves an average positioning error of 3.6 cm and an average yaw estimation error of 2.35°, significantly outperforming existing state-of-the-art acoustic localization and rigid-body pose estimation approaches.**

## I. INTRODUCTION

Indoor positioning technology provides essential information such as position and orientation for various robotic platforms. Depending on the task type and application requirements, commonly used indoor positioning approaches include wireless positioning technologies based on Bluetooth Low Energy (BLE) [1], Wi-Fi [2], along with vision-based positioning systems [3], and additionally acoustic positioning systems.

Acoustic indoor positioning systems have been widely studied, yielding various localization methods [4], most of which rely on accurate Time of Arrival (ToA) estimation. Representative approaches include Generalized Cross-Correlation (GCC) [5], [6], where GCC-PHAT achieves about 3 cm accuracy but is limited to short ranges [7], and Fractional Fourier Transform (FrFT)-based methods, which report approximately 30 cm accuracy at 30 m [8]. Inspired by millimeter-wave radar, Frequency-Modulated Continuous Wave (FMCW) techniques have also been adopted for ToA estimation [9], [10], achieving high ranging accuracy under ideal conditions and centimeter-level accuracy in constrained indoor environments [11], [12]. However, in realistic indoor scenarios, ToA estimation remains challenging: GCC-based methods are highly sensitive to multipath, while although FMCW techniques transform multipath effects into the frequency domain, reliably identifying the direct path remains difficult.

Meanwhile, due to the relatively low propagation speed of sound, acoustic positioning systems are highly susceptible to Doppler effects, which introduce biases into ToA and TDoA estimates. Although several Doppler compensation methods have been proposed, most can only partially mitigate this effect. In [13], [14], Doppler shifts are compensated by estimating the time–frequency slope variation of LFM signals using the FrFT, but such methods incur high computational complexity and depend on accurate slope estimation. Alternatively, up- and down-chirp signal designs have been adopted to counteract Doppler effects [15], [16]; however, Doppler-induced errors are not fully eliminated and additional signal bandwidth is required. Consequently, more robust Doppler compensation strategies are still needed for practical indoor scenarios.

In addition, rigid-body localization (RBL) has received increasing attention in recent years. AoA-based RBL methods achieve relatively high update rates but suffer from limited operating ranges [17]. Single-anchor acoustic approaches further simplify system deployment, yet their positioning accuracy remains at the sub-meter level [18], [19]. Other studies report angular errors of about 10° in confined environments [20], while two-anchor configurations achieve average heading errors of approximately 6.42° [21]. Overall, existing RBL methods face trade-offs among localization accuracy, operating range, and robustness, limiting their applicability in practical indoor scenarios. These limitations motivate the development of a more accurate and robust rigid-body localization approach for real-world environments.

To address the above issues, several key methodological improvements are introduced in this work. First, a FMCW-based first-path detection framework incorporating an early-arrival prior in the ToA estimation process is proposed to enhance direct-path identification under severe multipath conditions. Second, an up- and down-chirp signal design combined with a dedicated Doppler compensation algorithm is developed to effectively suppress motion-induced Doppler bias without increasing signal bandwidth or requiring additional hardware. Finally, to overcome the limited localization accuracy of conventional rigid-body identification methods, an inlier-selection-based rigid-body identification strategy is introduced, enabling more robust and accurate three-dimensional pose estimation using multi-microphone measurements.

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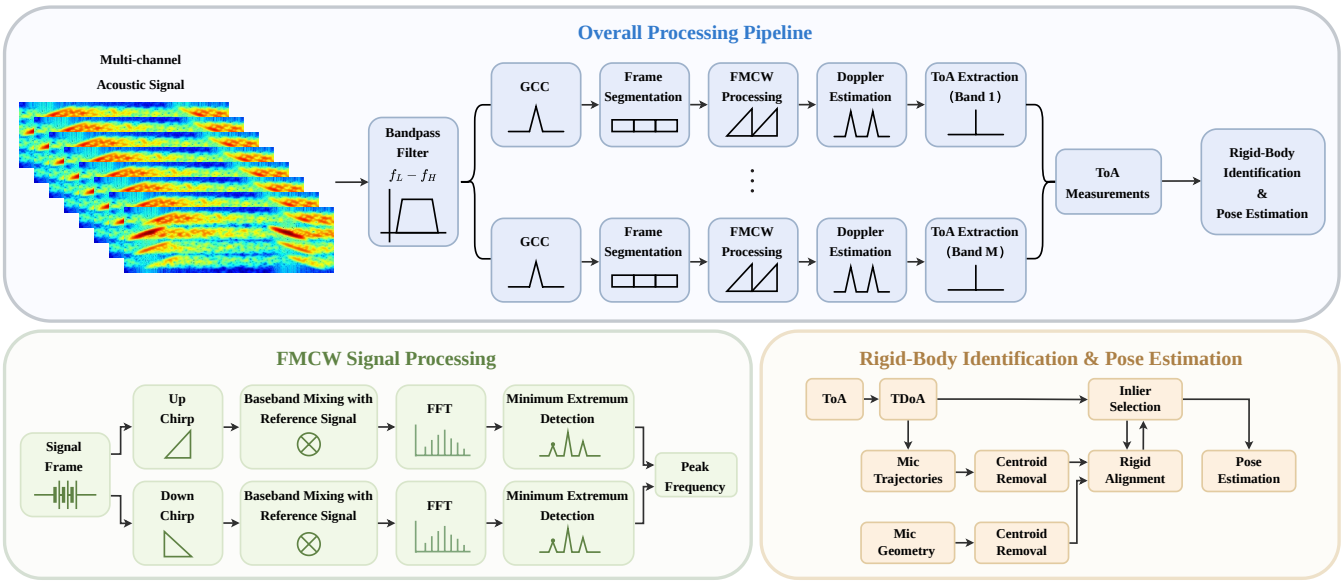



Fig. 1: Overview of the proposed acoustic localization framework.

Based on these improvements, we develop DRAPE (Doppler-Robust Acoustic Pose Estimation System), a Doppler-robust acoustic pose estimation framework for mobile robots in indoor environments. Comprehensive experiments are conducted in real indoor scenarios to evaluate the proposed system. The main contributions of this work can be summarized as follows:

- A FMCW-TDoA-based pose estimation framework integrating rigid-body constraints is developed for accurate three-dimensional pose estimation in indoor environments.
- A Doppler-shift elimination strategy based on up- and down-chirp signal design is proposed to mitigate motion-induced bias without additional bandwidth overhead.
- An inlier-selection-based rigid-body identification method is proposed to improve the robustness and accuracy of pose estimation using multi-microphone measurements.
- Extensive experiments in a $7.21\,\mathrm{m} \times 5.67\,\mathrm{m}$ indoor environment show that DRAPE achieves an average orientation error of $2.35^\circ$ and an average positioning error of 3.6 cm.

## II. DOPPLER-ROBUST ACOUSTIC POSE ESTIMATION SYSTEM

Fig. 1 illustrates the framework of the proposed acoustic positioning method, detailing the complete processing pipeline from multi-channel acoustic signal acquisition, preprocessing, and feature extraction to localization parameter estimation and three-dimensional pose estimation. The anchors continuously transmit specially designed LFM signals, which are received by the microphone array mounted on the Unmanned Ground Vehicle (UGV). The received signals are processed using an improved FMCW-TDoA method to estimate the ToA, while a Doppler compensation algorithm is applied to suppress motion-induced Doppler effects and obtain accurate TDoA measurements. Finally, a rigid-body identification strategy that exploits the geometric constraints of multiple microphones enables high-precision three-dimensional pose estimation of the UGV.

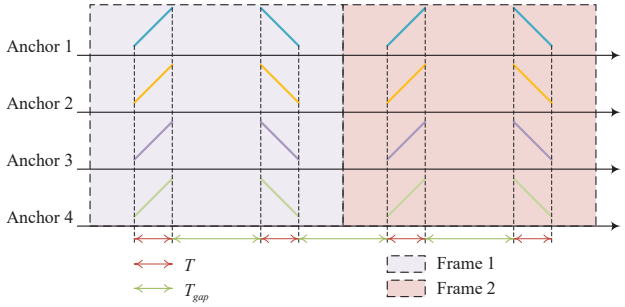



Fig. 2: Signal schedule of the proposed DRAPE system. The four acoustic anchors transmit up-chirp and down-chirp signals at distinct carrier frequencies. Each frame consists of one up-chirp followed by one down-chirp.

### A. Design of a Time-Division Dual-Chirp Signal

Unlike conventional acoustic localization systems, the proposed DRAPE system employs dual-chirp signals in the 15–23 kHz frequency band as the signal source. This frequency band is evenly divided into four sub-bands, each assigned to a fixed acoustic anchor. To reduce the bandwidth occupation of each anchor, an interleaved up- and down-chirp transmission scheme is adopted, in which a down-chirp signal is transmitted after a fixed and short time interval following the up-chirp transmission. Since this interval is sufficiently small, the position of the UGV can be assumed to remain unchanged during the transmission of a chirp pair. In contrast, the system in [22] adopts a continuous triangular-

wave transmission scheme and is mainly evaluated in small-scale environments. When the operating area is enlarged, acoustic reverberation may accumulate and raise the effective noise floor to an unacceptable level, which motivates the introduction of a guard interval between successive chirp transmissions in our design. The mathematical expressions of the designed dual-chirp signals are given as follows:

$$s^m_{\text{up}}(t) = \cos\left(2\pi\left(f^m_c t + \frac{B}{2T}t^2\right)\right), \tag{1}$$

$$s^m_{\text{down}}(t) = \cos\left(2\pi\left((f^m_c + B)t - \frac{B}{2T}t^2\right)\right). \tag{2}$$

where $s^m_{\text{up}}(t)$ and $s^m_{\text{down}}(t)$ denote the up-chirp and down-chirp signals transmitted by the $m$-th anchor, respectively. $T$ represents the duration of a single chirp, with $t \in [0, T]$. $B$ denotes the signal bandwidth, and $f^m_c$ is the starting frequency assigned to the $m$-th anchor.

The above dual-chirp signal model characterizes the frequency modulation structure of a single anchor within one transmission cycle. To provide a clearer illustration of how the signals from multiple anchors are organized in the time and frequency domains, Fig. 2 depicts the time–frequency structure of the proposed dual-chirp transmission scheme, including the chirp transmission order, the inter-chirp interval, and the frequency-band allocation among different anchors. In this work, the chirp duration is set to $T = 0.03\,\text{s}$ and the silent interval to $T_{\text{gap}} = 0.07\,\text{s}$ to mitigate multipath and reverberation effects in indoor environments.

### *B. Multipath-Robust FMCW First-Path Estimation*

Each individual chirp signal emitted by the anchor can be formulated as:

$$s(t) = \cos\left(2\pi\left(f_c t + \frac{B}{2T}t^2\right)\right). \tag{3}$$

In an ideal propagation environment without blockage or multipath effects, the signal transmitted by the anchor propagates over a distance $R$ and is received at the microphone as:

$$r(t) = \cos\left(2\pi\left(f_c(t-\tau_d) + \frac{B}{2T}(t-\tau_d)^2\right)\right), \tag{4}$$

where the propagation delay is given by $\tau_d = R/c$, and $c$ denotes the speed of sound in air.

In practical propagation environments, the received signal is often affected by multipath propagation, and the signal observed at the microphone can be expressed as:

$$s_r(t) = \sum_{\ell=0}^{L-1} \alpha_\ell \cos\left(2\pi\left(f_c(t-\tau_\ell) + \frac{B}{2T}(t-\tau_\ell)^2\right)\right), \tag{5}$$

where $\ell$ denotes the index of the propagation path, $L$ represents the total number of propagation paths, and $\alpha_\ell$ and $\tau_\ell$ denote the attenuation factor and propagation delay associated with the $\ell$-th propagation path, respectively.

To estimate the propagation delay between signals, a two-stage delay estimation method based on FMCW is proposed. Specifically, the GCC method is first employed to compute the cross-correlation between the received signal and the reference signal, and a coarse estimate of the propagation delay is obtained by locating the maximum peak of the correlation function:

$$\hat{\tau} = \arg\max_{\tau} \int s_r(t)\, s(t-\tau)\, \mathrm{d}t. \tag{6}$$

However, in practical environments with severe multipath effects, the GCC method often fails to accurately estimate the time of arrival due to the interference of reflected paths. Therefore, a secondary fine-search method based on FMCW is employed to further refine the delay estimation and enhance its accuracy and robustness.

Taking the initial delay estimate $\hat{\tau}$ as a reference, a signal segment with a duration of $T$ is extracted starting from $\hat{\tau}$. To ensure that the first-arriving (direct) path is included, the extraction window is further extended backward by a short duration denoted as $T_{\text{pre}}$, which is set to $T_{\text{pre}} = 0.008$ s in practice.

By mixing the transmitted signal with the received signal and applying a low-pass filter to the mixed signal, the intermediate-frequency (IF) signal is obtained:

$$r_{\text{IF}}(t) = \cos\left(2\pi\left(\frac{B}{T}\tau_d t + f_c\tau_d - \frac{B}{2T}\tau_d^2\right)\right). \tag{7}$$

Likewise, when multipath propagation is taken into account, the resulting IF signal can be written as:

$$s_{\text{IF}}(t) = \sum_{\ell=0}^{L-1} \frac{\alpha_\ell}{2} \cos\left(2\pi\left(\frac{B}{T}\tau_\ell t + f_c\tau_\ell - \frac{B}{2T}\tau_\ell^2\right)\right). \tag{8}$$

By applying the Fast Fourier Transform (FFT) to the IF signal, signal components corresponding to different propagation paths are transformed into distinct frequency components:

$$S_{\text{IF}}(\omega) \approx \sum_{\ell=0}^{L-1} A_\ell\, \delta\left(\omega - \frac{2\pi B}{T}\tau_\ell\right). \tag{9}$$

By selecting the minimum extremum in the frequency spectrum, the beat frequency corresponding to the direct path can be identified, thereby yielding the frequency of the first-arriving path:

$$\hat{f}_0 = \frac{1}{2\pi}\min\left\{\omega \;\middle|\; \omega \in \arg extremum\,|S_{\text{IF}}(\omega)|\right\}, \tag{10}$$

where $\arg extremum(\cdot)$ denotes the set of indices corresponding to the local maxima of the magnitude spectrum.

### *C. Doppler Shift Cancellation*

According to [9], $R + vt = c\tau_d$. By substituting this relationship into (7), we obtain:

$$r_{\text{IF}}(t) \approx \cos\left(2\pi\left(\frac{B}{cT}Rt + \frac{B}{cT}vt^2 + \frac{f_c}{c}R + \frac{f_c}{c}vt\right)\right), \tag{11}$$

where the second-order term proportional to $1/c^2$ has been neglected. This approximation is valid under the practical assumption that the target velocity is much smaller than the acoustic propagation speed and the observation interval is sufficiently short.

The instantaneous phase of the intermediate-frequency signal is extracted from the cosine argument and expressed as:

$$\phi(t) = 2\pi\left(\frac{B}{cT}Rt + \frac{B}{cT}vt^2 + \frac{f_c}{c}R + \frac{f_c}{c}vt\right). \tag{12}$$

The instantaneous frequency is then obtained by differentiating the instantaneous phase with respect to time:

$$f_{\mathrm{IF}}(t) = \frac{B}{cT}R + \frac{2B}{cT}vt + \frac{f_c}{c}v. \tag{13}$$

Within a single chirp period, the time variable $t$ can be approximated by its average value $T/2$. Accordingly, the intermediate-frequency $f_{\mathrm{IF}}$ can be written as:

$$f_{\mathrm{IF}} = \frac{B}{cT}R + \frac{B}{c}v + \frac{f_c}{c}v. \tag{14}$$

Accordingly, different expressions can be derived for the instantaneous frequencies corresponding to the up-chirp and down-chirp signals:

$$f_{\mathrm{IF}}^{\mathrm{up}} = \frac{B}{cT}R + \frac{B}{c}v + \frac{f_c}{c}v, \tag{15}$$

$$f_{\mathrm{IF}}^{\mathrm{down}} = \frac{B}{cT}R + \frac{B}{c}v - \frac{f_c + B}{c}v. \tag{16}$$

After obtaining the estimated frequency $\hat{f}$ from (10), the corresponding observation model can be expressed as

$$\begin{bmatrix} \hat{f}_{\mathrm{up}} \\ \hat{f}_{\mathrm{down}} \end{bmatrix} = \begin{bmatrix} \frac{B}{cT} & \frac{B+f_c}{c} \\ \frac{B}{cT} & \frac{-f_c}{c} \end{bmatrix} \begin{bmatrix} R \\ v \end{bmatrix} + \varepsilon, \tag{17}$$

where $\varepsilon$ denotes the frequency estimation error.

Since the radial velocity $v$ mainly acts as a Doppler-induced nuisance parameter and is physically bounded in indoor environments, while the distance $R$ is of primary interest, a velocity-regularized estimation formulation is adopted.

Specifically, the estimation of $R$ and $v$ is formulated as the following optimization problem:

$$\min_{R,v} \left\| \begin{bmatrix} \frac{B}{cT} & \frac{B+f_c}{c} \\ \frac{B}{cT} & \frac{-f_c}{c} \end{bmatrix} \begin{bmatrix} R \\ v \end{bmatrix} - \begin{bmatrix} \hat{f}_{\mathrm{up}} \\ \hat{f}_{\mathrm{down}} \end{bmatrix} \right\|_2^2 + \lambda v^2, \tag{18}$$

where $\|\cdot\|_2$ denotes the $\ell_2$ norm, and $\lambda$ is a regularization parameter.

This formulation corresponds to a MAP estimator with a zero-mean Gaussian prior on the radial velocity, which suppresses Doppler-induced bias and improves the robustness of distance estimation. The corresponding ToA estimates for the up- and down-chirp signals are then derived from the estimated distance $R$:

$$t_{\mathrm{i}}^{m,\mathrm{up}} = \hat{\tau}_{\mathrm{i}}^{m,\mathrm{up}} - T_{\mathrm{pre}} + \frac{R_{\mathrm{i}}^m}{c}, \tag{19}$$

$$t_{\mathrm{i}}^{m,\mathrm{down}} = \hat{\tau}_{\mathrm{i}}^{m,\mathrm{down}} - T_{\mathrm{pre}} + \frac{R_{\mathrm{i}}^m}{c}, \tag{20}$$

where the subscript $i$ denotes the microphone index, and the superscript $m$ represents the index of the acoustic anchor. The superscripts "up" and "down" correspond to the up-chirp and down-chirp signals, respectively.

**Algorithm 1:** 3D Pose Estimation

**Input:** Body-frame microphone positions $\mathbf{P}_b \in \mathbb{R}^{2\times N}$; Global-frame microphone positions $\mathbf{P}^k \in \mathbb{R}^{2\times N}$; Number of frames $K$; Initial threshold $\tau$; Maximum iterations $I_{\max}$

**Output:** Yaw angle $\psi^{(k)}$ and translation vector $\mathbf{t}^{(k)}$

**for** $k = 1$ **to** $K$ **do**
- Center global positions $\bar{\mathbf{P}}^k \leftarrow \mathbf{P}^k - \mathrm{mean}(\mathbf{P}^k)$
- Initialize inlier set $\mathscr{I} \leftarrow \{1, 2, \ldots, N\}$
- **for** $i = 1$ **to** $I_{\max}$ **do**
  - Compute correlation matrix $\mathbf{H} \leftarrow \mathbf{P}_b(:, \mathscr{I})\bar{\mathbf{P}}^k(:, \mathscr{I})^T$
  - Perform SVD $\mathbf{H} = \mathbf{U}\Sigma\mathbf{V}^T$
  - Estimate rotation $\mathbf{R} \leftarrow \mathbf{V}\mathbf{U}^T$
  - **if** $\det(\mathbf{R}) < 0$ **then**
    - Correct reflection $\mathbf{V}(:, \mathrm{end}) \leftarrow -\mathbf{V}(:, \mathrm{end})$, $\mathbf{R} \leftarrow \mathbf{V}\mathbf{U}^T$
  - **end**
  - Align body-frame points $\hat{\mathbf{P}} \leftarrow \mathbf{R}\mathbf{P}_b$
  - Compute residuals $e_j \leftarrow \|\hat{\mathbf{P}}_j - \bar{\mathbf{P}}_j^k\|_2, \ \forall j \in \mathscr{I}$
  - Compute adaptive threshold $\mu \leftarrow \mathrm{median}(e)$, $d \leftarrow \mathrm{median}(|e - \mu|)$, $\tau_{\mathrm{adp}} \leftarrow \max(\tau, \mu + 2.5d)$
  - Update inlier set $\mathscr{I}_{\mathrm{new}} \leftarrow \{j \mid e_j < \tau_{\mathrm{adp}}\}$
  - **if** $\mathscr{I}_{\mathrm{new}} = \mathscr{I}$ **then**
    - **break**
  - **end**
  - $\mathscr{I} \leftarrow \mathscr{I}_{\mathrm{new}}$
- **end**
- Compute centroids $\mathbf{c} \leftarrow \mathrm{mean}(\mathbf{P}^k(:, \mathscr{I}))$, $\mathbf{c}_b \leftarrow \mathrm{mean}(\mathbf{P}_b(:, \mathscr{I}))$
- Estimate translation $\mathbf{t}^k \leftarrow \mathbf{c} - \mathbf{R}\mathbf{c}_b$
- Extract yaw angle $\psi^k \leftarrow \mathrm{atan2}(R_{21}, R_{11})$

**end**

The actual ToA is obtained by averaging the ToA estimates derived from the up- and down-chirp signals:

$$t_i^m = \frac{1}{2}\left(t_i^{m,\mathrm{up}} + t_i^{m,\mathrm{down}}\right). \tag{21}$$

Therefore, the final TDoA estimate can be expressed as:

$$\Delta t_i^{m1} = t_i^m - t_i^1. \tag{22}$$

### *D. Rigid-Body 3D Pose Estimation*

Based on the obtained TDoA measurements, the spatial coordinates of the microphones can be determined using a hyperbolic localization method. Although the microphone positions are ultimately estimated in the two-dimensional horizontal plane, the full three-dimensional coordinates of the anchors are incorporated into the localization model to mitigate errors caused by height mismatches between the microphones and the anchors.

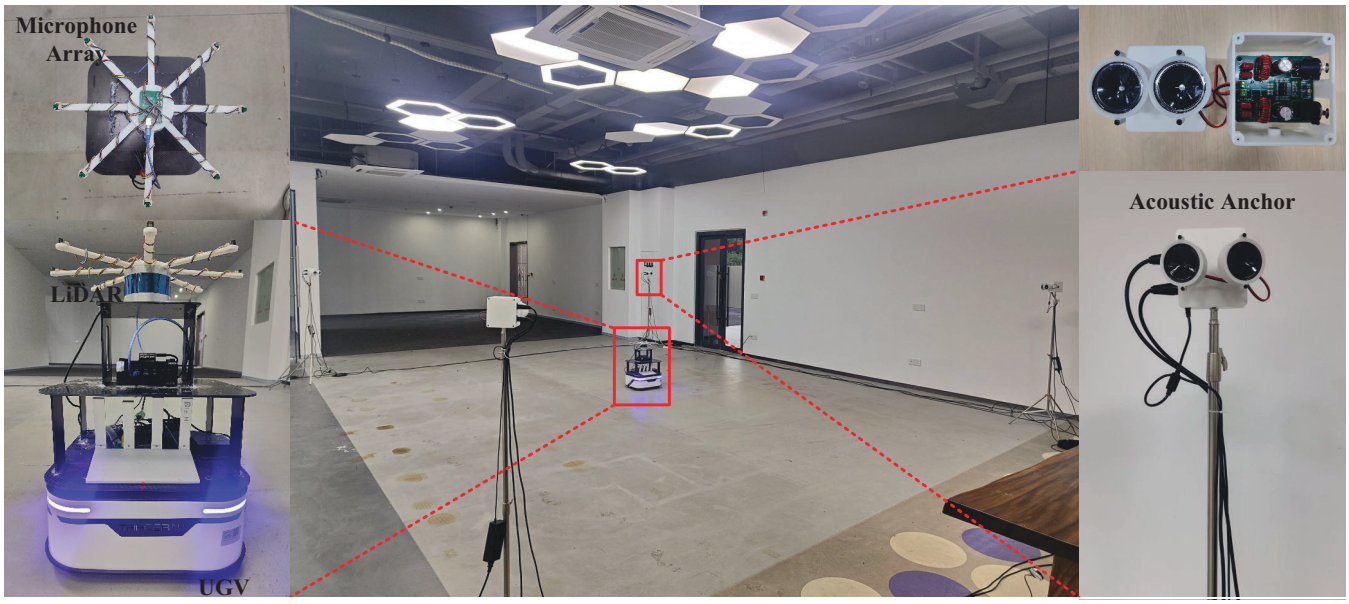



Fig. 3: Spatial layout of the experimental environment and associated equipment. Four custom-built acoustic anchors are deployed in the environment, with full time synchronization among all anchors. An eight-microphone array mounted on the UGV is used to capture the acoustic signals transmitted by the anchors.

Let $\mathbf{q}_m = [u_m,\ v_m,\ w_m]^T$ denote the known three-dimensional position of the $m$-th anchor, and let $\mathbf{p}_i = [x_i,\ y_i,\ z_i]^T$ denote the position of the $i$-th microphone, where the microphone height $z_i$ is known *a priori*. The acoustic propagation distance between the microphone and the $m$-th anchor is defined as

$$d_{im} = \|\mathbf{p}_i - \mathbf{q}_m\|. \tag{23}$$

Using anchor 1 as the reference, the TDoA measurements give rise to the following hyperbolic constraints:

$$d_{im} - d_{i1} = c\,\Delta t_i^{m1}, \quad m = 2,\ldots,M. \tag{24}$$

Since the microphone height $z_i$ and the anchor heights $w_m$ are known, the propagation distance can be explicitly expressed as

$$d_{im} = \sqrt{(x_i - u_m)^2 + (y_i - v_m)^2 + (z_i - w_m)^2}. \tag{25}$$

By collecting the residuals corresponding to all anchors $m = 2,\ldots,M$, the TDoA localization problem can be formulated in a vector form as

$$\mathbf{f}(x_i, y_i) = \begin{bmatrix} f_2(x_i, y_i) \\ f_3(x_i, y_i) \\ \vdots \\ f_M(x_i, y_i) \end{bmatrix}, \tag{26}$$

where each residual term is defined by

$$f_m(x_i, y_i) = d_{im} - d_{i1} - c\,\Delta t_i^{m1}. \tag{27}$$

The planar microphone position is then estimated by solving the following nonlinear least-squares problem:

$$\hat{\mathbf{p}}_i = \arg\min_{x_i, y_i} \sum_{m=2}^{M} f_m(x_i, y_i)^2, \tag{28}$$

where $\hat{\mathbf{p}}_i = [x_i,\ y_i]^T$.

To enhance robustness against outliers caused by measurement noise and localization errors, an iterative rigid-body alignment method with inlier selection is employed to jointly estimate the overall pose of the microphone array. Specifically, at each time step $k$, the microphone position matrix $\mathbf{P}^k$ in the global coordinate frame is constructed by stacking the planar microphone position estimates $\hat{\mathbf{p}}_i$ obtained from the TDoA-based localization. Due to the presence of noise in the TDoA measurements and potential instability of the nonlinear solver, these frame-wise position estimates may contain outliers. Therefore, the rigid-body alignment procedure performs iterative optimization over the set of $\mathbf{P}^k$ across multiple time frames, simultaneously estimating the array pose and rejecting inconsistent measurements. The overall procedure is summarized in Algorithm 1.

## III. EXPERIMENTAL SETUP

To validate the effectiveness and robustness of the proposed DRAPE, systematic TDoA-based acoustic localization and pose estimation experiments were conducted using a complete indoor experimental platform. The experimental environment and anchor layout are shown in Fig. 3, and the overall setup, including the environment, hardware configuration, and system parameters, is described below.

Four acoustic anchors were deployed in a $7.21\,\mathrm{m} \times 5.67\,\mathrm{m}$ area, each transmitting chirp signals in a dedicated ultrasonic sub-band (15–17 kHz, 17–19 kHz, 19–21 kHz, and 21–23 kHz). The transmission schedule is shown in Fig. 2. All anchors were time-synchronized to ensure reliable TDoA measurements.

An eight-channel microphone array was mounted on the top of a UGV to receive the acoustic signals emitted by

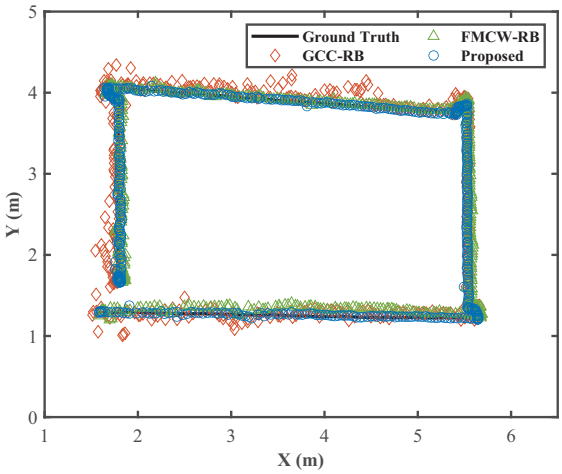



Fig. 4: Comparison of the trajectories estimated by different methods with the ground truth. To avoid visual clutter and improve visualization, the positions of the acoustic base stations are not shown in the figure.

the anchors. The microphones were uniformly arranged on a circular array with a radius of 20 cm, and the audio signals were synchronously sampled at a rate of 48 kHz. The experiments were conducted using an AgileX Robotics TRACER MINI mobile robot as the UGV platform.

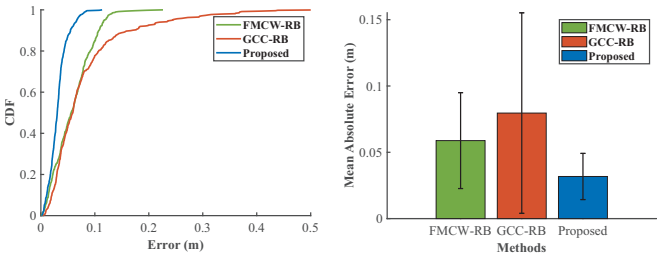
For ground-truth acquisition, a Velodyne VLP-16 LiDAR was rigidly mounted on the UGV to provide accurate pose measurements throughout the experiments. A global reference frame was defined based on the experimental environment, with fixed relative transformations between the LiDAR and microphone array coordinate frames. All estimated microphone positions and array poses were expressed in this global frame for evaluation. Meanwhile, multi-channel acoustic data and LiDAR measurements were time-stamped and recorded using a unified data acquisition system, ensuring channel-level synchronization and temporal alignment for offline processing.

## IV. RESULTS AND DISCUSSION

Building upon the experimental setup and data collection described in the previous section, the performance of the proposed approach is systematically evaluated using data obtained from real indoor experiments. The evaluation uses high-precision ground-truth poses provided by the LiDAR system and focuses on three key aspects: position estimation accuracy, yaw angle estimation accuracy, and anti-Doppler performance under different velocities. For comparison, GCC-RB and FMCW-RB are adopted as baseline methods, where RB denotes the proposed rigid-body identification strategy. Trajectory comparisons, statistical error analysis, and cumulative distribution functions (CDFs) are employed to provide a quantitative assessment of the proposed method.

### A. Position Estimation Performance

As shown in Fig. 4, the estimated trajectories obtained using GCC-RB, FMCW-RB, and the proposed DRAPE are




(a) (b)

Fig. 5: Comparison of position estimation errors for different estimation methods: (a) CDF of the position estimation errors; (b) statistical comparison of the position estimation errors.

TABLE I: Comparison of position and yaw angle estimation accuracy

| Metric | Method | MAE | RMSE | Median / 90% / Max |
|---|---|---|---|---|
| Position | GCC-RB | 8.0 | 11.0 | 5.9 / 17.0 / 52.9 |
| | FMCW-RB | 5.9 | 6.9 | 5.6 / 10.8 / 22.6 |
| | DRAPE | **3.2** | **3.6** | **3.0 / 5.5 / 11.3** |
| Yaw | GCC-RB | 7.1 | 14.7 | 4.6 / 12.9 / 126.6 |
| | FMCW-RB | 3.6 | 5.0 | 2.7 / 7.0 / 20.8 |
| | DRAPE | **1.6** | **2.3** | **1.1 / 3.8 / 8.4** |

*Note: Position errors are measured in centimeters, and yaw angle errors are measured in degrees. Median / 90% / Max denote the median error, the 90th percentile error, and the maximum error, respectively.*

compared against the ground truth. It can be observed that the trajectory estimated by the GCC-RB method exhibits severe fluctuations, indicating poor stability and limited localization accuracy, with a position RMSE of 11.0 cm. In contrast, the FMCW-RB method is able to more reliably identify the first propagation path, thereby partially mitigating multipath interference and improving localization accuracy, achieving an RMSE of 6.9 cm. Building upon this improvement, the proposed DRAPE method further enhances trajectory estimation performance, reducing the position RMSE to 3.6 cm while producing significantly smoother and more consistent trajectories.

Based on the qualitative trajectory comparison, a statistical evaluation is further conducted to quantitatively analyze the position estimation accuracy and stability of different methods. Fig. 5 presents both the CDFs of the position estimation errors and the corresponding error bar statistics, which together characterize the error distribution and dispersion. It can be seen that the CDF curve of DRAPE consistently lies above those of the comparison methods over the entire error range, indicating superior localization accuracy. Meanwhile, the error bar results show that DRAPE achieves a lower mean positioning error with reduced variance, demonstrating improved accuracy and robustness.

Specifically, when the error threshold is set to 5 cm, approximately 87.4% of the position estimates obtained by DRAPE fall within this bound, whereas the corresponding proportions for FMCW-RB and GCC-RB are 45.3% and 43.7%, respectively. This observation is consistent with the

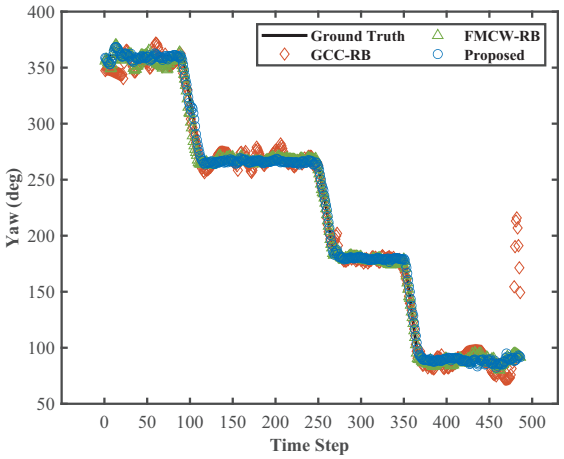
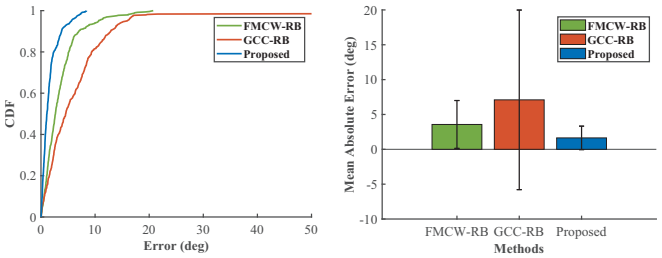



Fig. 6: Comparison of yaw angle estimates obtained by different methods with the ground truth.




(a) (b)

Fig. 7: Comparison of yaw angle estimation errors for different estimation methods: (a) CDF of the yaw angle estimation errors; (b) statistical comparison of the yaw angle estimation errors.

trajectory comparison results, where significant error fluctuations are evident for GCC-RB and, to a lesser extent, for FMCW-RB, while DRAPE exhibits much smoother and more accurate trajectories. This performance gain is primarily attributed to the explicit suppression of Doppler-induced frequency bias in DRAPE, which effectively decouples range estimation from target motion and prevents systematic ToA deviations under non-zero radial velocity. As a result, DRAPE significantly suppresses outliers and reduces the occurrence of large localization errors.

In addition, the statistical position error metrics of different methods are summarized in Table I, including the MAE, RMSE, median error, 90% error, and maximum error. It can be observed that DRAPE achieves the best performance across all reported metrics, with MAE, RMSE, and median error values of 3.2 cm, 3.6 cm, and 3.0 cm, respectively, which are significantly lower than those of FMCW-RB and GCC-RB. Moreover, the 90% error and maximum error of DRAPE are substantially smaller, indicating not only higher overall localization accuracy but also stronger robustness against large errors and outliers. These statistical results are consistent with the trajectory comparison and CDF analysis, further validating the effectiveness and reliability of the proposed method in real indoor environments.

### *B. Yaw Angle Estimation Performance*

Fig. 6 compares the yaw angle estimation results obtained using the GCC-RB method, the FMCW-RB method, and the proposed DRAPE approach against the ground truth. The GCC-RB method suffers from pronounced temporal jitter and occasional abrupt deviations in yaw estimation, leading to limited stability and relatively large angular errors, with a yaw RMSE of 14.7°. By more reliably isolating the dominant propagation path, the FMCW-RB method achieves improved angular smoothness and reduces the yaw estimation error to an RMSE of 4.95°. Among all evaluated methods, the proposed DRAPE approach delivers the most accurate and stable yaw estimation performance, attaining the lowest RMSE of 2.35°.

Beyond the qualitative trajectory comparison, a comprehensive quantitative evaluation is conducted to further characterize the statistical properties of the yaw angle estimation errors. Fig. 7 presents both the CDFs and the corresponding error bar statistics for different methods, offering complementary insights into the overall error distribution and estimation variability. Compared with GCC-RB and FMCW-RB, DRAPE exhibits a markedly steeper CDF curve together with smaller error bars, indicating that a larger proportion of yaw estimates are concentrated within a narrow error range while maintaining reduced variance and enhanced robustness.

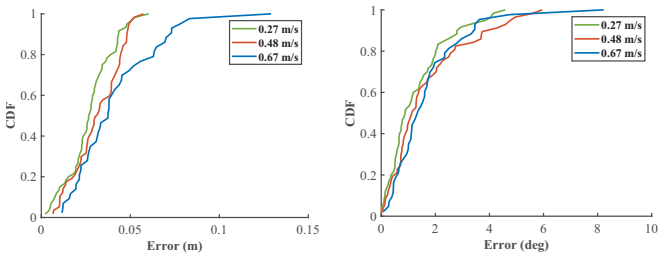
To further substantiate these observations, Table I summarizes key statistical error metrics, including the MAE, RMSE, median error, 90% error, and maximum error. These metrics jointly reflect both the average estimation accuracy and the sensitivity of each method to outliers and extreme deviations. Consistent with the CDF and error bar results, DRAPE achieves the lowest errors across all reported metrics. This performance gain is primarily attributed to the explicit mitigation of Doppler-induced frequency bias, which effectively suppresses systematic angular estimation errors under target motion. Consequently, DRAPE demonstrates superior robustness and reliability in yaw angle estimation under practical experimental conditions.

### *C. Performance Evaluation of Anti-Doppler Estimation*

Fig. 8 shows the CDFs measured under different velocity conditions. It can be observed that the proposed algorithm maintains stable estimation performance across varying velocities, indicating its robustness against Doppler effects.

For position estimation, at a velocity of 0.27 m/s, the median position error is approximately 2.7 cm, with 63% of the samples exhibiting errors below 3 cm. As the velocity increases to 0.48 m/s, the median position error remains at 3.2 cm, and 95% of the errors are bounded within 5 cm, showing only marginal degradation. Even at the highest tested velocity of 0.67 m/s, the proposed approach achieves a median position error of 3.8 cm, with 72% of the samples below 5 cm, demonstrating robust position estimation under stronger Doppler effects.

In terms of yaw angle estimation, similar trends can be observed across different velocities. At 0.27 m/s, the median




(a) (b)

Fig. 8: Estimation error comparison under different velocities: (a) CDF of position estimation errors; (b) CDF of yaw angle estimation errors.

yaw error is 0.9°, and 65% of the errors fall below 1.5°. When the velocity increases to 0.48 m/s, the median yaw error remains at 1.1°, with 68% of the samples constrained within 2°. At 0.67 m/s, the median yaw error is 1.4°, and more than 79% of the errors are bounded within 2.5°, indicating that yaw angle estimation accuracy is largely preserved despite increasing Doppler effects.

Overall, these results demonstrate that the proposed algorithm effectively mitigates the impact of velocity-induced Doppler distortions, enabling reliable position and yaw angle estimation across a wide range of motion speeds.

## V. CONCLUSIONS

This work presents an improved FMCW-TDoA-based framework for accurate three-dimensional pose estimation of mobile robots in indoor environments. A novel Doppler-shift elimination strategy is introduced to suppress Doppler-induced bias in acoustic measurements without additional bandwidth or hardware modifications. By incorporating rigid-body constraints of the microphone array, the proposed DRAPE system achieves robust and stable pose estimation under practical motion conditions.

Extensive experiments conducted in a 7.21 m × 5.67 m indoor environment demonstrate the effectiveness of the proposed approach. Results at different motion velocities confirm the robustness of the method against Doppler effects. The DRAPE system achieves an average orientation error of 2.35° and an average positioning error of 3.6 cm, consistently outperforming representative state-of-the-art methods.

Future work will focus on increasing the system refresh rate while further improving pose estimation accuracy and robustness, enabling more reliable and stable performance in dynamic and challenging indoor environments.

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